

JEE MAIN REVISION MASTER: LAWS OF MOTION

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 152 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Newton's Laws & Free Body Diagrams	95 PYQs	Dynamics
Very High	Friction (Static, Limiting & Kinetic)	62 PYQs	Friction

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Newton's Laws & Free Body Diagrams	$\Sigma \vec{F} = m\vec{a}$, $T = \frac{2m_1 m_2}{m_1 + m_2}g$, $a = \frac{m_2 - m_1}{m_1 + m_2}g$
Friction (Static, Limiting & Kinetic)	$f_{\text{max}} = \mu_s N$, $f_k = \mu_k N$, $\theta_{\text{repose}} = \tan^{-1} \mu_s$, $a = g(\sin \theta - \mu_k \cos \theta)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Newton's Laws & Free Body Diagrams (Very High Yield — 95 PYQs)

Core Principle: Equations of motion for connected bodies, string-pulley constraints, and apparent weight in non-inertial frames.

■ **Caution / JEE Pitfall:** In an accelerating elevator with upward acceleration a , apparent weight is $N = m(g + a)$, NOT $m(g - a)$.

■ **Speed Trick:** Use string constraint $\Sigma T_i \cdot x_i = 0 \implies \Sigma T_i \cdot a_i = 0$ to relate accelerations in complex pulley systems.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionl..." [Ans: B]
- [2026 Morning Shift] Q#2 (MCQ): "Three masses $m_{(1)}=4\text{kg}$, $m_{(2)}=4\text{kg}$ and $m_{(3)}=6\text{kg}$ are suspended from a fixed smooth frictionless pully as shown in the figure below. The va..." [Ans: A]
- [2026 Morning Shift] Q#2 (MCQ): "A 4 kg mass moves under the influence of a force $\vec{F}=(4t^3\hat{i}-3t\hat{j})\text{N}$ where t is the time in second. If mass starts from o..." [Ans: C]

3.2 Friction (Static, Limiting & Kinetic) (Very High Yield — 62 PYQs)

Core Principle: Static self-adjusting friction up to limiting value $\mu_s N$ and constant kinetic friction $\mu_k N$ on rough surfaces.

■ **Caution / JEE Pitfall:** Static friction is self-adjusting: $f_s = F_{\text{(applied)}}$ as long as $F_{\text{(applied)}} \leq \mu_s N$. It does NOT automatically equal $\mu_s N$.

■ **Speed Trick:** In two-block systems, find the maximum common acceleration allowed by friction before determining if slipping occurs.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless..." **[Ans: B]**
- **[2026 Evening Shift] Q#1 (MCQ):** "A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with co..." **[Ans: B]**
- **[2026 Evening Shift] Q#1 (MCQ):** "A block is sliding down on an inclined plane of slope θ and at an instant $t=0$ this block is given an upward momentum so that it sta..." **[Ans: C]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pulley which is fixed on the edge of a cube (S) , as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ . The cube (S) moves towards the right with acceleration of $(g / 2)$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is ____ (take $g=10\text{m/s}^2$)

- (A) 0.33
- (B) 0.67 **[Correct]**
- (C) 1
- (D) 0.5

Correct Answer: B

Solution Analysis: Since the cube is accelerating at $a=(g / 2)$ to the right, when analysing the blocks from the cube's non-inertial frame, we must apply a pseudo-force ($F_{\text{(p)}}=m \times a$) on both blocks in the opposite direction to the acceleration of block S. For block P (Mass $m_{\text{(P)}}=2\text{kg}$) Along horizontal the tension (T) in the thread pulls it to the right. The pseudo-force ($m_{\text{(P)}} \cdot a$) pulls it to the left. Along vertical direction gravity ($m_{\text{(P)}}g$) acts downward and normal force ($N_{\text{(P)}}$) from the top surface acts upward. Since the block is stationary relative to the cube : $T=m_{\text{(P)}} \cdot a$

Benchmark Problem 2: JEE Main 2026 (Online) 28th January Evening Shift (Q#1)

A small block of mass m slides down from the top of a frictionless inclined surface, while the inclined plane is moving towards left with constant acceleration a_0 . The angle between the inclined plane and ground is θ and its base length is L . Assuming that initially the small block is at the top of the inclined plane, the time it takes to reach the lowest point of the inclined plane is _____.

- (A) $\sqrt{\frac{2L}{g \sin \theta - a_0 \cos \theta}}$
- (B) $\sqrt{\frac{4L}{g \sin 2\theta - a_0(1 + \cos 2\theta)}}$ **[Correct]**
- (C) $\sqrt{\frac{2L}{g \sin 2\theta - a_0(1 + \cos 2\theta)}}$
- (D) $\sqrt{\frac{4L}{g \cos^2 \theta - a_0 \sin \theta \cos \theta}}$

Correct Answer: B

Solution Analysis: The length of the incline (hypotenuse) is given by : $\cos \theta = (L / s) \Rightarrow s = (L / \cos \theta)$ Since the wedge is accelerating to the left with a_0 , we must work in the non-inertial frame of the wedge. This causes a pseudo-force (ma_0) acting on the block to the right. We resolve all forces along the surface of the incline : The net acceleration relative to the incline (a_{rel}) is : $a_{rel} = \frac{mg \sin \theta - ma_0 \cos \theta}{m} = g \sin \theta - a_0 \cos \theta$ Using the kinematic equation for distance s starting from rest ($u=0$) : $s = \frac{1}{2} a_{rel} t^2$. Putting